

Step 5: Testing and Refinement

Test Case 1: Scheduled Feeding, Pet Eats

- **Input:** Current time = 8:00 AM, food available, manual feed not pressed.
- **Expected Output:** Servo rotates for 3 seconds, bowl weight decreases $\geq 5g$, no alert sent.
- **Actual Output:** Servo rotates, bowl weight decreased by 6g, no alert sent.
- **Result:** Pass. Feeding logic works correctly.

Test Case 2: Scheduled Feeding, Pet Does Not Eat

- **Input:** Current time = 6:00 PM, food available, manual feed not pressed.
- **Expected Output:** Servo rotates, bowl weight decreases $< 5g$, "Uneaten Food" alert sent.
- **Actual Output:** Servo rotates, bowl weight decreased by 2g, alert sent.
- **Result:** Pass. Alert mechanism works.

Test Case 3: System Fails to Detect Uneaten Food in Small Portions

- **Input:** Scheduled feeding at 8:00 AM, food available, pet eats only 2g (less than threshold of 5g).
- **Expected Output:** Servo rotates, bowl weight decreases $< 5g$, "Uneaten Food" alert sent.
- **Actual Output:** Servo rotates, bowl weight decreased by 2g, no alert sent (system only flags weight decrease $\leq 5g$, but small portions may be eaten unnoticed).
- **Result:** Fail. System fails to notify owner about partial feeding.

Test Case 4: Servo Malfunction & Food Not Dispensed

- **Input:** Scheduled feeding at 6:00 PM, food available, manual feed not pressed.
- **Expected Output:** Servo rotates, food dispensed, bowl weight decreases $\geq 5g$, no alert sent.
- **Actual Output:** Servo fails to rotate due to motor malfunction, bowl weight unchanged, no food dispensed, no alert for mechanical failure.
- **Result:** Fail. Feeding did not occur, and system did not detect servo failure.

Test Case 5: Manual Feeding Request

- **Input:** Manual feed button pressed, food available.
- **Expected Output:** Servo rotates, bowl weight decreases $\geq 5g$, no alert.
- **Actual Output:** Servo rotates, bowl weight decreased by 5g, no alert.
- **Result:** Pass. Manual feeding works as intended.

Test Case 6: Food Bin Empty

- **Input:** Current time = 8:00 AM, food bin empty.
- **Expected Output:** No servo action, "Refill Food" alert sent.
- **Actual Output:** No servo action, alert sent.
- **Result:** Pass. Food availability logic works.

Discussion of Logic Issue

The current system uses a fixed threshold of 5g to determine whether food has been eaten. This threshold is too high for small pets or situations where the pet consumes only a partial portion. Consequently, the system may incorrectly assume that food remains uneaten, which can lead to missed alerts or inaccurate feeding records.

Suggested Refinements

- **Dynamic Threshold:** Adjust the weight threshold based on the pet's size or the portion size to more accurately detect partial feeding.
- **Multiple Monitoring Intervals:** Monitor the bowl weight at several points in time (e.g., every 5 minutes over a 30-minute period) to capture slow or partial consumption.
- **Visual or Sensor Confirmation:** Integrate a camera or motion sensor to visually confirm food consumption, providing an additional layer of verification.